Experiment No: 12

LOAD TEST ON 3-PHASE SLIP RING INDUCTION MOTOR

Aim:

Performing Brake test on a 3-phase Slip Ring Induction Motor and obtaining its performance characteristics.

Machine details: 3KW, 3 Φ, 415 V 50 Hz, 1410 rpm

Instruments Required:

S.No	Name of Instruments	Specifications
1.	Ammeter	
2.	Voltmeter	
3.	Wattmeter	
4.	Tachometer	
5.	3-Φ autotransformer	

< students are expected to calculate instrument ranges based on the machine ratings before start of experiment and get it approved before connections are made>

Principle:

Slip ring induction motor is also called as phase wound motor. The motor is wound for as many poles as the no. of stator poles and always wound 3-Φ even while the stator is wound two-phase. The other three windings are brought out and connected to three insulated slip-rings mounted on the shaft with brushes resting on them. These three brushes are further externally connected to a three phase star connected rheostat. This makes possible the introduction of an additional resistance in the rotor circuit during starting period for increasing starting torque of the motor. On load, the slip enables the required rotor current to be circulated, as demanded by the load.

Procedure:

(a)Initial settings: Initially the TPST switch is at open position. The external resistance in the rotor circuit should be kept at max. value. The machine must be started at no load.

(b) Execution of the Experiment: Connections are given as per the circuit diagram and the motor is started by using Rotor resistance starter at no load. As speed increases, the external resistance is gradually cut out. Then the no load meter readings are noted. Then the motor is loaded step by step till we get the rated current and the readings of the voltmeter, ammeter, wattmeters, spring balance are noted.

Result:

The performance characteristics of a 3-Φ slip ring induction motor by conducting load test has been drawn.

<Give some specific results here>

Inference: < Students are expected to write an inference based on the results obtained >